SK

PORTFOLIO · 2026

Siphiwe Khumalo

Building automation, IT infrastructure and software, treated as one system.

IT Administrator at Spartcon Technologies. I develop building management interfaces on Tridium Niagara 4, set up the infrastructure and remote access behind them, and build internal business software.



2 BMS deployments
ARC GLASSHOUSE · DOH MIDDELBURG

Niagara 4 Certified
TRIDIUM N4 TECHNICAL CERT.

3 disciplines
AUTOMATION · IT · SOFTWARE

01 / PROFILE

I like knowing what is actually happening underneath a system, and then making it dependable.

I'm **Siphiwe Khumalo**, known to most people as Ronnie. I started in freelance web development and hands-on IT support. That mix taught me that problems rarely stay in one layer: a "broken dashboard" can be a network route, a controller setting or a single point name.

Since joining **Spartcon Technologies** as IT Administrator in August 2026, I have worked on live building automation projects. I developed the operator interfaces for the ARC Glasshouse and Department of Health (Middelburg) BMS deployments, set up the site infrastructure and remote access behind them, and built internal business applications for the team.

I'm still early in my career, and I treat each deployment as a chance to deepen my engineering knowledge. In October 2026 I completed Tridium's five-day Niagara 4 training and passed the exam with 83%, earning the Niagara 4 Technical Certification.



Siphiwe "Ronnie" Khumalo

APPROACH 01

Start where the fault really is

Trace a problem through hardware, network, controller and software before deciding what to change.

APPROACH 02

Make the safe action the easy one

Confirm every command, verify it against the controller, and never present old data as live.

APPROACH 03

Hand over something people can run

Clear interfaces, written guides and on-site training, so the system keeps working after I leave.

AT A GLANCE

ROLE

IT Administrator, Spartcon Technologies, since August 2026

FOCUS

Building automation · IT infrastructure · Software development

MILESTONES

ARC handover 9 Oct 2026 · Niagara 4 certified 30 Oct 2026 · Middelburg handover 11 Nov 2026

02 / EXPERIENCE

Professional experience

AUG 2026 – PRESENT
FULL TIME

IT Administrator

Spartcon Technologies

- **Building automation.** Developed the operator interfaces for BMS deployments at ARC Glasshouse and the Department of Health facility in Middelburg, working with Tridium Niagara 4 and JACE controllers.
- **IT infrastructure.** Set up and maintained the site host PC that runs the Middelburg dashboard around the clock, including start-up automation and data backup.
- **Networking and connectivity.** Configured site connectivity and secure remote access so the client can monitor the facility from off site.
- **Technical troubleshooting.** Diagnosed communication, point-mapping and control issues during commissioning, together with the controls integrator.
- **Software solutions.** Designed and built internal applications for Spartcon, including Spartcon Capture (invoice processing) and Spartcon Timesheet.
- **Deployment and handover.** Produced user guides and signage, trained client staff on site and supported formal project handover.

BEFORE AUG 2026
FREELANCE

Web Developer & IT Support

Independent

- **Web development.** Built websites, landing pages and interactive tools for small businesses and personal projects, using HTML, CSS, JavaScript and third-party APIs.
- **IT support.** Hands-on hardware, network and systems troubleshooting, including Microsoft 365 and general user support.

2026 MILESTONES

01 Aug

Joined Spartcon Technologies as IT Administrator.

09 Oct

ARC Glasshouse BMS handed over.

30 Oct

Niagara 4 Technical Certification (Tridium), exam score 83%.

11 Nov

Department of Health Middelburg BMS handed over, staff trained.

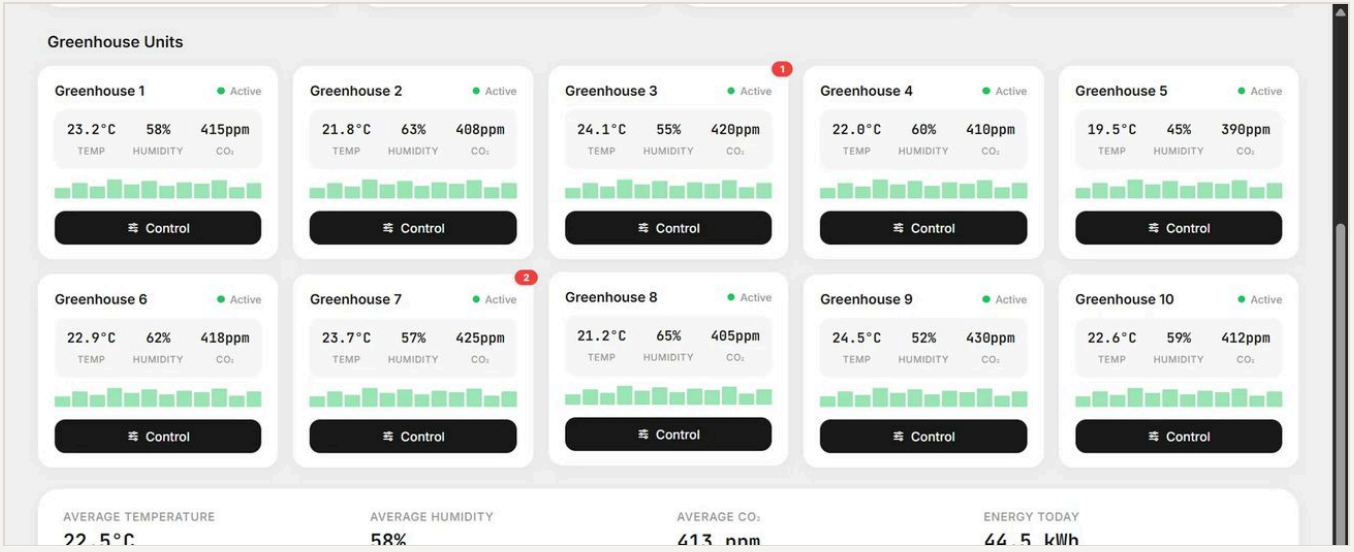
01

BUILDING MANAGEMENT SYSTEM · SOFTWARE

ARC Glasshouse: greenhouse BMS interface

A custom operator interface for the Agricultural Research Council's glasshouse facility. It monitors and controls ten greenhouses from wall-mounted touchscreens, running on a Tridium Niagara 4 controls platform.

CLIENT	DELIVERED WITH	MY ROLE	HANDOVER
ARC Glasshouse	Spartcon Technologies	BMS interface software	9 October 2026



Greenhouse overview: ten units with temperature, humidity, CO_2 and short trends.

OVERVIEW

The glasshouse is made up of ten research greenhouses. Each one needs climate conditions and irrigation that staff can see and adjust quickly, from screens on the floor rather than an engineering workstation.

OBJECTIVES

- Show temperature, humidity and CO_2 for every greenhouse at a glance.
- Give operators control of HVAC and irrigation schedules from touchscreens.
- Run reliably on the ten deployed Windows PoE displays.

MY ROLE

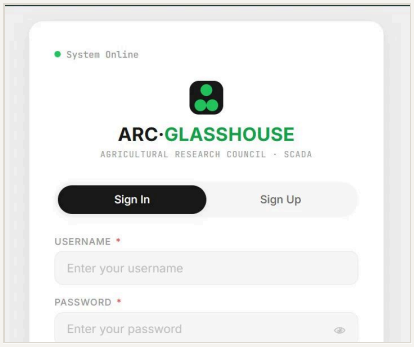
- Developed the custom BMS user interface.
- Designed the HVAC monitoring and control graphics.
- Implemented the interface for ten Windows PoE touchscreen displays.
- Worked within the Niagara 4 environment.
- Prepared and refined the software for the deployment environment.
- Supported interface testing, troubleshooting and final handover.

TECHNOLOGY & EQUIPMENT

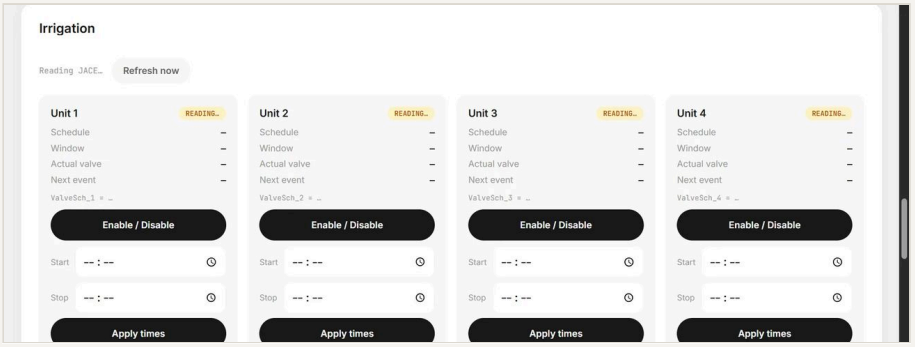
`Tridium Niagara 4` `JACE controller` `BACnet/IP` `HTML · CSS · JavaScript` `10 × Windows PoE touchscreens`

IMPLEMENTATION

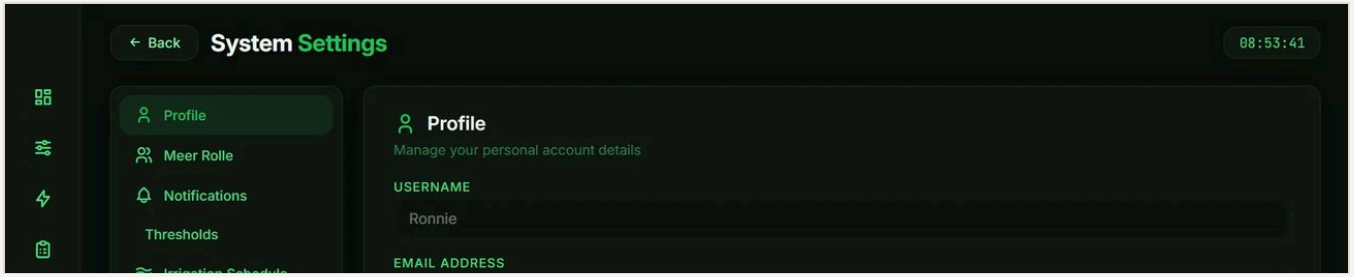
The interface is a touch-first web application that reads and writes points on the Niagara station. It provides a **greenhouse overview** with live readings and short trends, a **control view** for each unit, **irrigation scheduling** for ten valve schedules with start and stop times, and **system settings** for user roles, notification thresholds and device configuration, behind a sign-in screen.



Secure sign-in.



Irrigation scheduling for each valve.



System settings: roles, thresholds, irrigation and device configuration (personal details blurred).

CHALLENGES & PROBLEM-SOLVING

CHALLENGE	APPROACH
Point names in the station did not always match the documentation, which made the irrigation schedules hard to bind correctly.	Confirmed the exact slot names on site against the Workbench configuration, then bound each of the ten valve schedules to its verified point.
The touchscreens are used standing, at a distance, often during other work.	Large controls, high-contrast readings and one clear action per button, sized for touch rather than a mouse.
Operators must never mistake an unanswered request for a real value.	Every value shows a clear "Reading…" state until the controller responds.

OUTCOME

The interface was deployed across the facility's ten touchscreen displays, and the project was handed over on 9 October 2026.

> **Scope:** my responsibility was the software, meaning the BMS interface and its deployment. The hardware, electrical and infrastructure work on this project was carried out by other project participants.

SCREENSHOTS FROM DEVELOPMENT AND TESTING. SOME VALUES ARE TEST DATA.

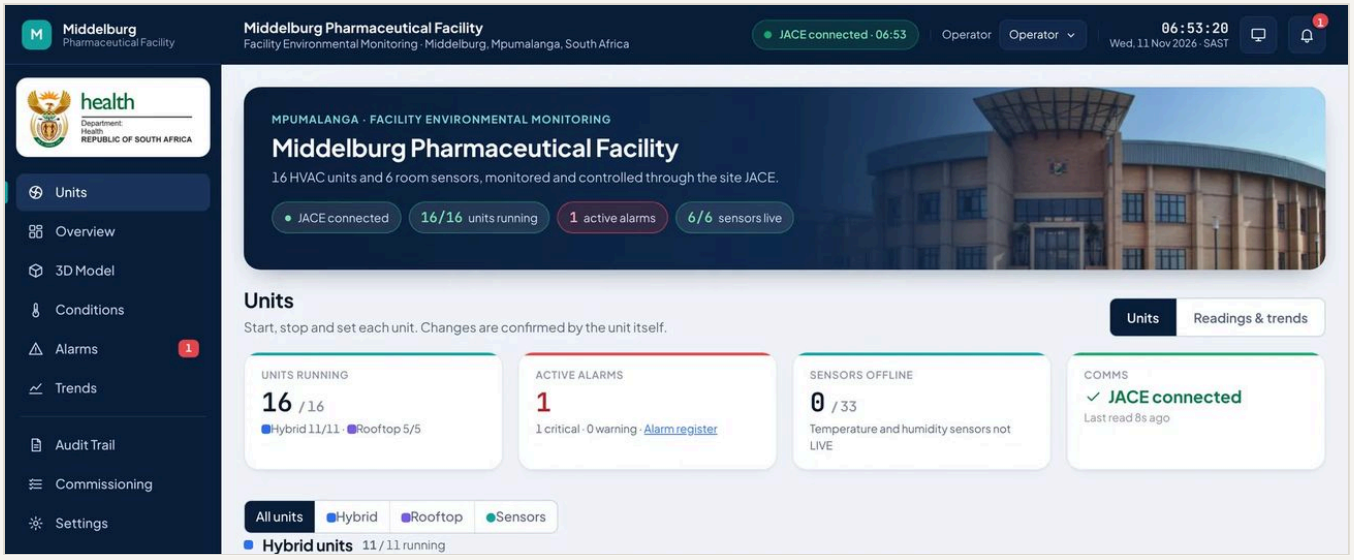
02

BUILDING MANAGEMENT SYSTEM · SOFTWARE · INFRASTRUCTURE

Department of Health, Middelburg: pharmaceutical facility BMS

A live monitoring and control dashboard for a pharmaceutical warehouse. It covers 16 HVAC units, 6 room sensors and 6 fire dampers connected through a Tridium JACE, running 24/7 on site and reachable securely from off site.

CLIENT	DELIVERED WITH	MY ROLE	HANDOVER
Department of Health	Spartcon Technologies	Software, infrastructure, handover	11 November 2026



Units page: site summary and a live card for every HVAC unit.

OVERVIEW

Medicine storage depends on stable temperature and humidity. The facility has 5 rooftop units, 11 hybrid units, 6 room sensors and 6 fire dampers. Staff needed one clear place to see conditions, act on alarms and adjust the rooftop units, without opening engineering tools.

OBJECTIVES

- Live visibility of every unit, zone and sensor.
- Safe start/stop and setpoint control for the rooftop units.
- Clear alarms, including dirty-filter warnings.
- Recorded history that can be exported for reporting.
- Access on site and securely from off site.

MY ROLE

- Designed and built the dashboard and its Node.js gateway.
- Worked with the controls integrator on point mapping and commissioning.
- Set up the site host PC to run the system 24/7 and start automatically.
- Configured connectivity and secure remote access.
- Wrote the operator user guide and cabinet signage, and trained staff at handover.

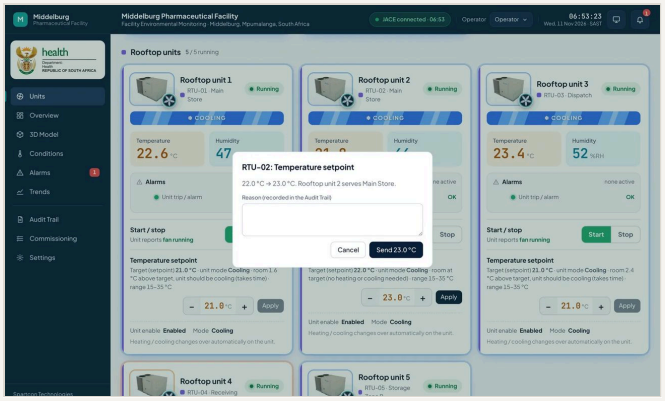
TECHNOLOGY & EQUIPMENT

`Tridium Niagara 4` `JACE controller` `BACnet · Modbus TCP` `Node.js` `JavaScript` `Windows host PC` `TP-Link Omada` `Tailscale`

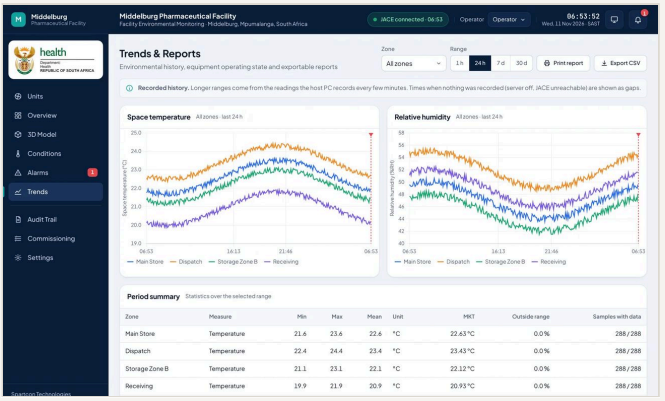
DASHBOARD SCREENSHOTS FROM A TEST ENVIRONMENT WITH SAMPLE VALUES.

IMPLEMENTATION

A small gateway on the host PC holds the controller credentials, so browsers never see them. Only an approved list of points can be written to. Every command needs confirmation and is **verified by reading the value back** from the controller. A shared cache and command queue protect the JACE from multiple viewers, and every point is logged every five minutes for trends and CSV export. The dashboard has Units, Overview, 3D Model, Alarms, Trends and Audit Trail views, with Viewer, Operator and Admin roles.



Every command is confirmed, then verified against the controller.



Trends: 24-hour history with period summary and MKT.

CHALLENGES & PROBLEM-SOLVING

CHALLENGE

The rooftop units would not switch between heating and cooling automatically. Leftover high-priority commands in the controller's priority array were overriding the changeover.

APPROACH

Traced the fault to the priority array, made heating/cooling read-only in the dashboard, and documented how the integrator should release the overrides.

CHALLENGE

During commissioning some points were not yet mapped, and others used different names on site, including the fire damper points.

APPROACH

Wrote diagnostic scripts that probe the controller for each expected point, then corrected the mapping on site with the integrator.

CHALLENGE

The site has no public IP address, but the client wanted to monitor it remotely.

APPROACH

Set up an encrypted private network (Tailscale) to the host PC, with a step-by-step access guide for client staff.

CHALLENGE

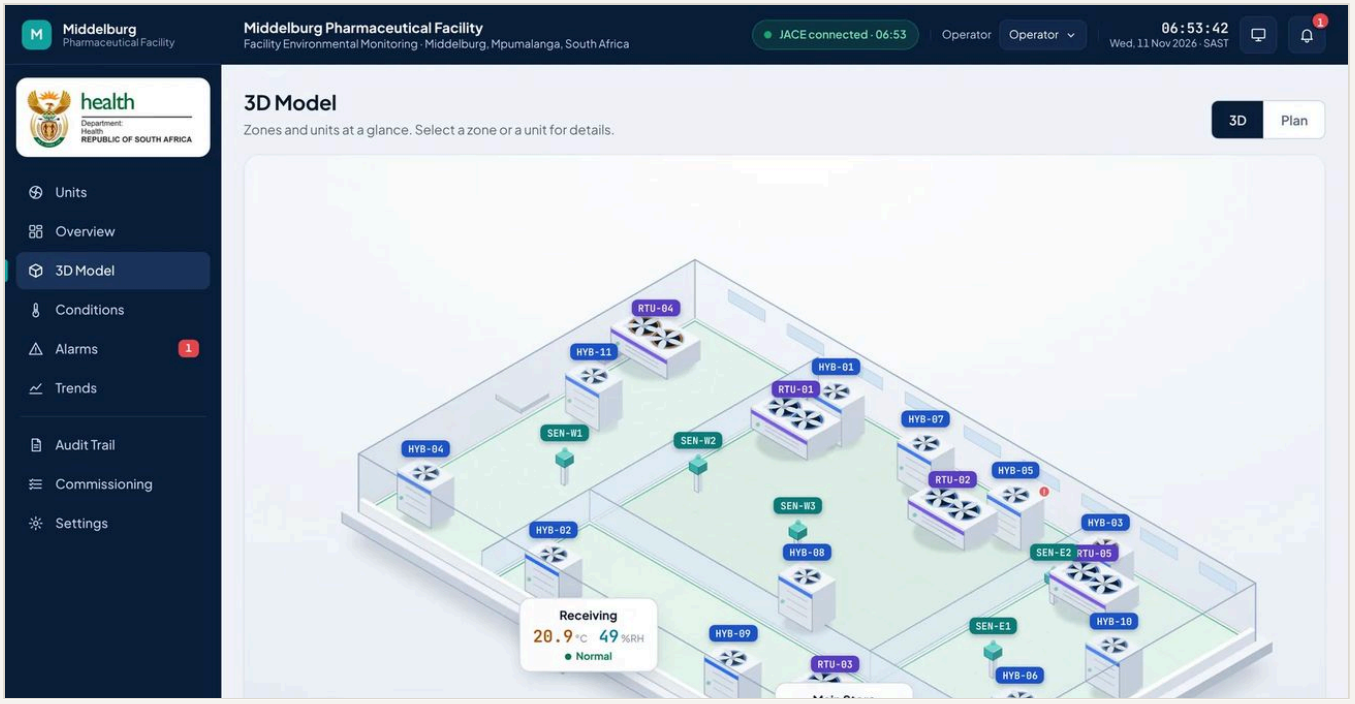
Values from a lost connection could be mistaken for live data.

APPROACH

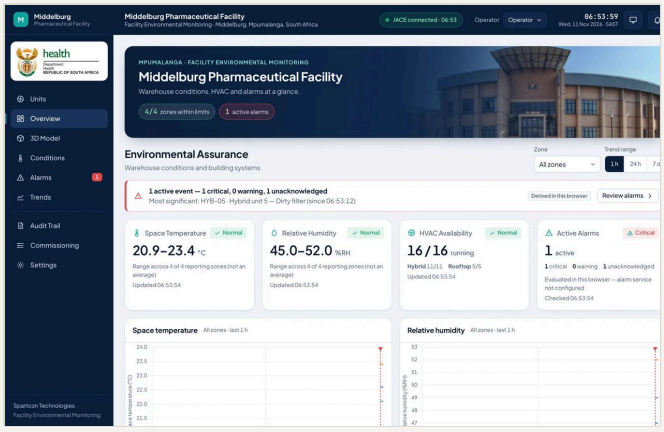
Every reading shows its state (LIVE, STALE or OFFLINE), and controls lock while the connection is down.

OUTCOME

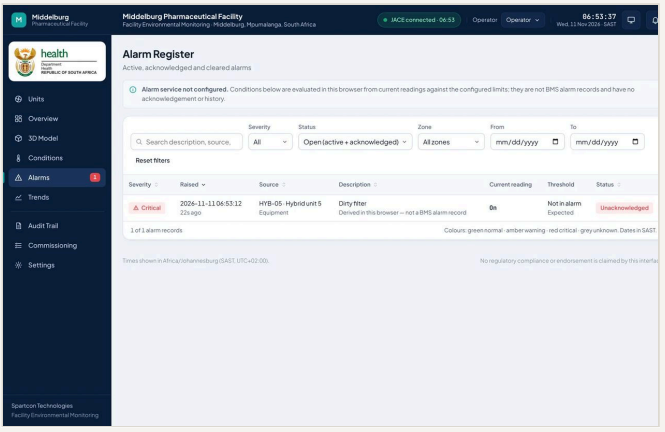
The system was handed over to the Department of Health on 11 November 2026, with staff trained using a 10-page illustrated user guide. It runs continuously on site and records every reading for later export.



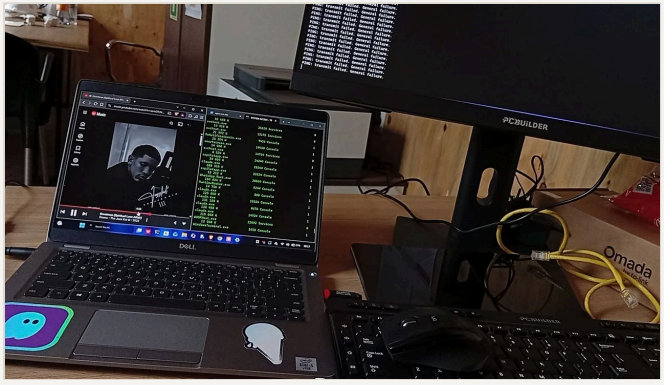
3D model: zones and units coloured by status. Select a zone or unit for details.



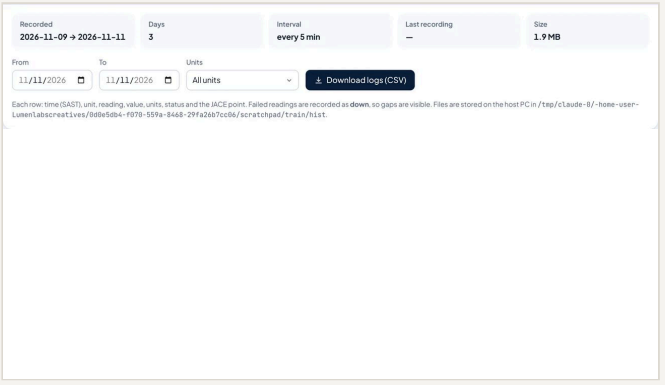
Overview: environmental assurance at a glance.



Alarm register for the whole site.



Commissioning: connectivity testing during the network setup.



Stored logs: export recorded data for any date range.

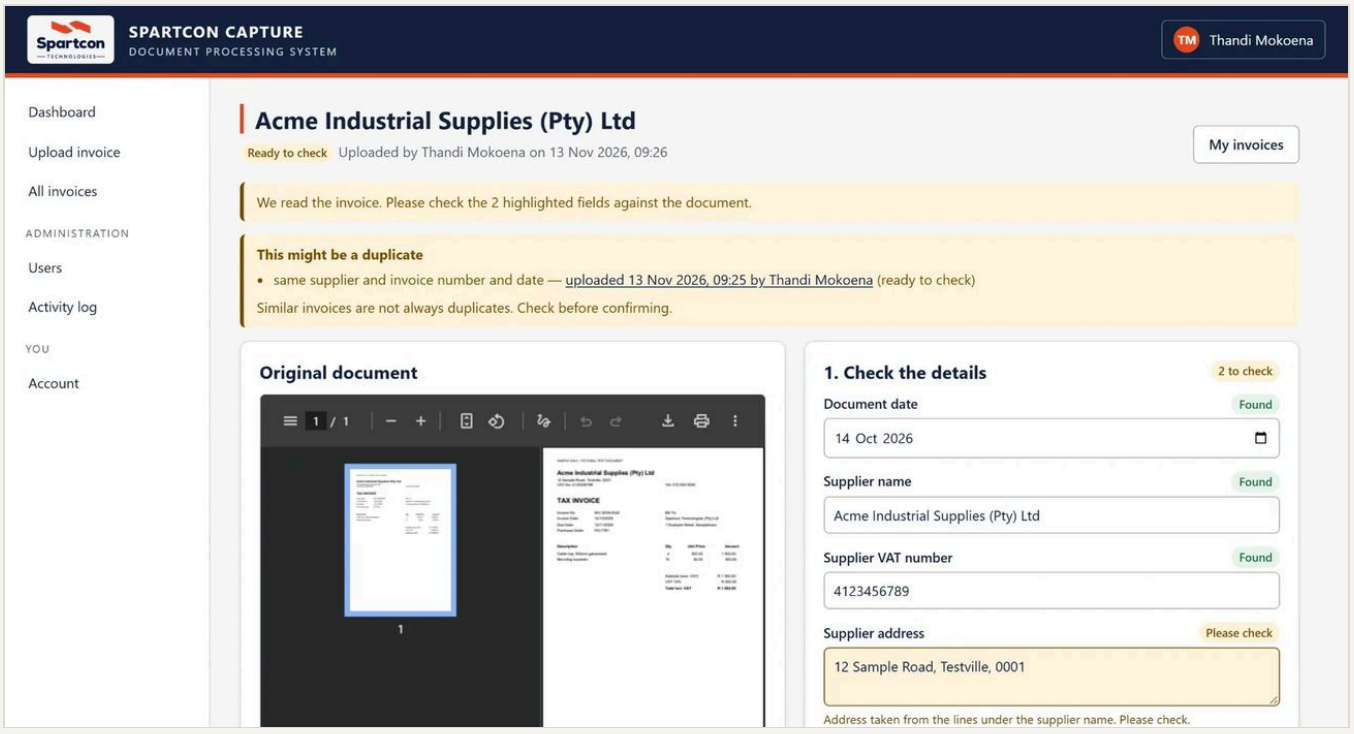
03

BUSINESS SOFTWARE · INTERNAL APPLICATION

Spartcon Capture: document processing system

An internal web application that turns supplier invoices and receipts, whether PDFs, scans or phone photos, into checked, structured records, with a full audit trail.

ORGANISATION	TYPE	MY ROLE	YEAR
Spartcon Technologies	Internal web application	Design & development	2026



Review: the original document beside the extracted fields, with a duplicate warning.

OVERVIEW

Capturing invoice details by hand is slow and error-prone. Spartcon Capture reads each uploaded document and fills in the key fields. A person still checks the result before anything is confirmed.

OBJECTIVES

- Cut manual typing of supplier, date, VAT number, invoice number and totals.
- Keep a person in control of every confirmed record.
- Catch duplicate invoices before they are processed twice.
- Record who did what, and when, in a tamper-evident log.

IMPLEMENTATION

- Upload a PDF or photo. Text-based PDFs are read directly, while scans and photos go through OCR.
- A review screen shows the original document beside the extracted fields. Each field is marked **Found** or **Please check**.
- A duplicate warning appears for matching supplier, invoice number and date.
- Confirmed documents produce an Excel file, and invoices move from *Ready to check* to *Done*.
- User administration and an append-only activity log with a log integrity check.

TECHNOLOGY

`Web application` `PDF text extraction` `OCR` `Excel export` `Role-based access` `Audit logging`

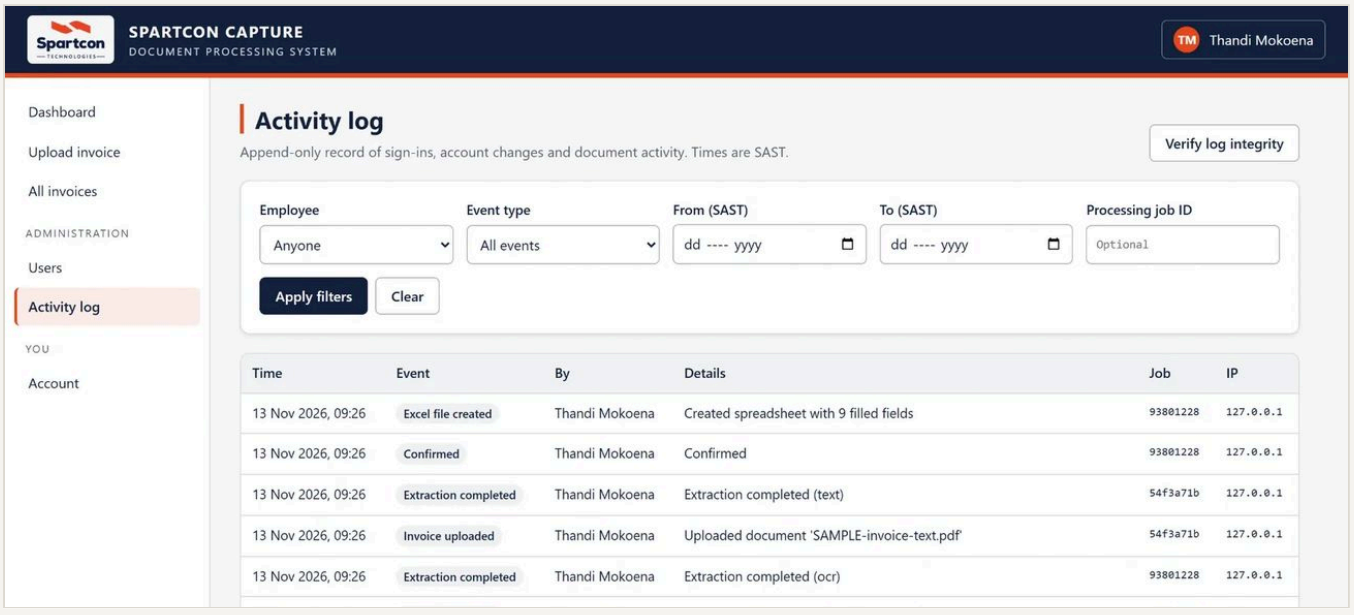
CHALLENGES & PROBLEM-SOLVING

Extraction is never perfect. Instead of trusting it blindly, the interface highlights exactly which fields need a human check, and why. For example, an address taken from the lines under the supplier name is flagged for review.

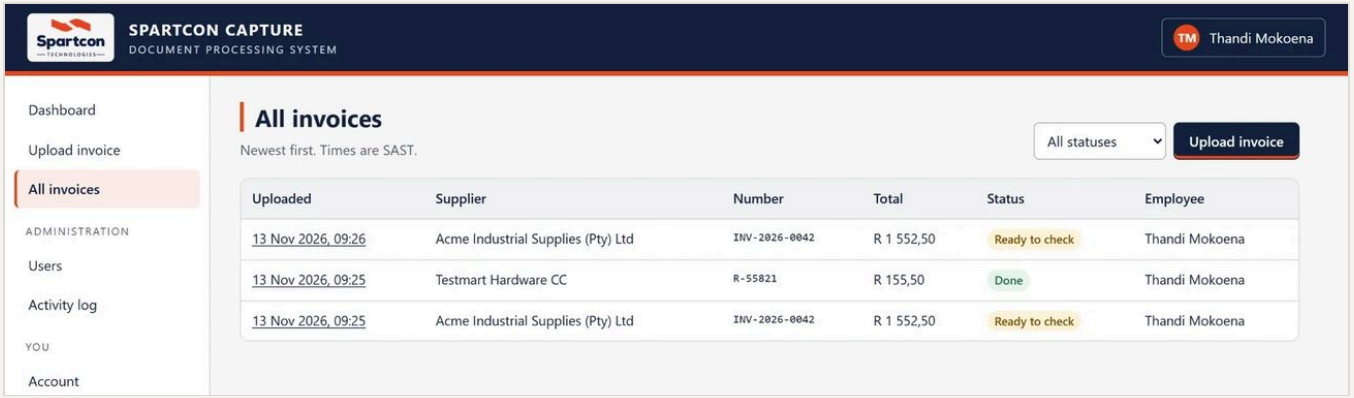
Accountability matters for finance data. Every sign-in, upload, extraction and confirmation is written to an append-only log, which can be verified for integrity.

OUTCOME

A complete invoice workflow, from upload and extraction through review, duplicate detection and export, with an audit trail at every step.



Append-only activity log with filters and integrity verification.



All invoices, newest first, with status filter.

04 / MORE WORK

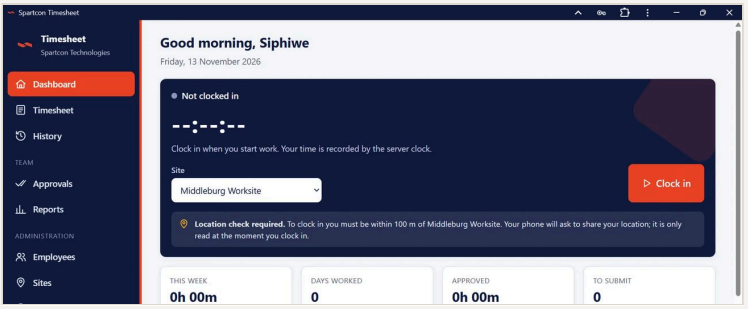
Additional Software & Web Projects

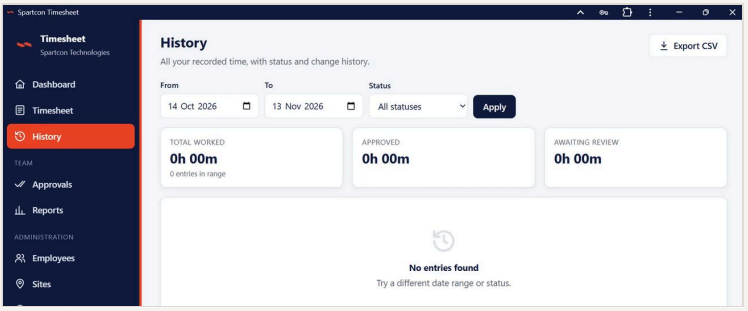
BUSINESS APPLICATION · SPARTCON

Spartcon Timesheet

A time and attendance app for site and office staff. Clocking in uses the server clock and a location check against the selected worksite.

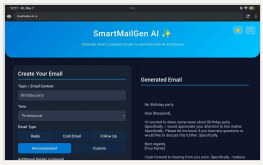
- Clock in and out per site, within 100 m of the worksite.
- Weekly totals, approvals and reports for team leads.
- History by date range and status, with CSV export.
- Employees, sites, location log and audit log.





Dashboard and history views.

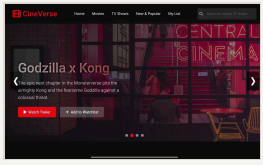
EARLIER WEB PROJECTS



SMARTMAILGEN

Generates context-aware email replies by connecting a simple interface to the OpenAI API. HTML · CSS · JavaScript.

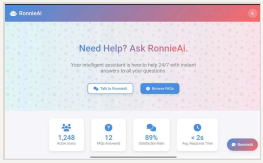
siphiwe-khumalo.github.io/Smart-Email-Generator



CINEVERSE

Movie discovery with live data fetching, filtering and recommendations from the TMDB API.

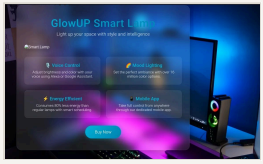
siphiwe-khumalo.github.io/CineVerse.html



RONNIEAI FAQ ASSISTANT

A conversational FAQ assistant that answers repetitive support questions.

siphiwe-khumalo.github.io/not-so-Smart-Ai-FAQs-Assistant



GLOWUP LAMP LANDING PAGE

A product landing page with an interactive showcase and a deliberate conversion flow.

siphiwe-khumalo.github.io/Glowup-Product-Landing-Page-

05 / SKILLS

Technical skills

Grouped by discipline. Each item is something I have used on real work.

01

Building Automation

- Tridium Niagara 4 and JACE controllers
- BMS operator interfaces and graphics
- HVAC monitoring and control
- Points, priority arrays and overrides
- Alarm and trend presentation

02

Networking & Infrastructure

- Router configuration (TP-Link Omada)
- LAN setup and connectivity testing
- Secure remote access (Tailscale)
- Site host PC deployment
- CCTV and VoIP fundamentals

03

IT Systems & Cloud

- Microsoft 365 administration and support
- Windows setup and administration
- Hardware installation and diagnostics
- User support and documentation
- Cybersecurity fundamentals

04

Software Development

- JavaScript, HTML and CSS
- Node.js and REST APIs
- React and Firebase
- Progressive Web Apps
- Git and GitHub

05

Systems Integration

- Web apps connected to building controllers
- Read-back verified control commands
- Data logging and CSV / Excel export
- Document extraction and OCR workflows
- Role-based access and audit trails

06

Troubleshooting & Deployment

- On-site commissioning support
- Fault-finding across network and controller
- Diagnostic scripting
- User guides and site signage
- Operator training and handover

06 / CERTIFICATIONS & TRAINING

Certifications & training

TRIDIUM · NIAGARA 4

Niagara 4 Technical Certification

Certificate of Completion from Tridium for completing all training and testing requirements of the five-day Niagara 4 course.

83%

TRAINING	26–30 October 2026 (five days)
CERTIFIED	30 October 2026
AUTH. ID	438218109
CEU	4.0



TRIDIUM CERTIFICATE OF COMPLETION

ADDITIONAL EDUCATION

Tourism studies. They shaped how I think about technology from the user's side.

07 / CONTACT

Let's talk about your system.

I'm open to conversations about building automation, IT infrastructure and software projects, and to roles where those disciplines meet.

Email	siphiwek@spartcongroup.co.za
LinkedIn	linkedin.com/in/siphiwe-khumalo-97a0aa306
GitHub	github.com/Siphiwe-Khumalo
Website	siphiwe-khumalo.github.io/Portfolio-Website-